

End-to-End Abnormal Subgraph Detection via Subgraph-Level Contrastive Learning

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Abstract—Abnormal subgraph (AS) detection plays a significant role in ensuring the security of many high-impact domains. Unlike node anomaly detection, identifying subgraph anomalies is extremely challenging due to the exponentially large subgraph space caused by various combinations of nodes and edges. Moreover, in the absence of supervisory signals, how to quantify the abnormality of subgraphs poses another pressing challenge. Traditional methods typically rely on handcrafted subgraph anomaly measures, making it hard to handle potential unknown anomalies with limited prior knowledge. Recent deep learning-based techniques are predominantly designed to discover individual node anomalies, which could be suboptimal for AS detection due to the inconsideration of collaborative behaviors between nodes in the subgraph. In fact, existing studies have put very little effort into this task, and even dedicated performance evaluation metrics are not yet available. To address the above challenges and promote related research, in this article, we propose an end-to-end unsupervised subgraph anomaly detection framework (EndSubG), which jointly models subgraph partition and AS detection as a whole instead of treating them as two separate stages. Specifically, EndSubG uncovers potential AS boundaries that violate the Homophily assumption by modeling the edge existence probability, then achieves anomaly-aware graph embedding and subgraph partition based on the refined topology. By forming a coarsened subgraph network, EndSubG picks out subgraph anomalies by learning the “subgraph-vicinity” matching patterns. Additionally, we design an evaluation metric weighted normalized mutual information centered on AS (AS-WNMI) specifically

for subgraph anomaly detection, which is a variant of vanilla NMI and quantifies detection performance from both subgraph partition and anomaly recognition. The experimental results on synthetic and real-world datasets corroborate the superiority of end-to-end unsupervised subgraph anomaly detection framework (EndSubG) in terms of area under the curve (AUC), average precision (AP), and AS-WNMI. We also provide an intuitive analysis of the detected subgraphs through visualization for better understanding.

Index Terms—Anomaly-aware graph embedding, contrastive learning, graph neural networks, subgraph anomaly detection.

I. INTRODUCTION

GRAPH-STRUCTURED data is ubiquitous and has proved to be a powerful tool to model and analyze various modern information networks, such as social networks, transaction networks, and biological networks. To prevent unpredictable hazards, graph anomaly detection has been widely used to help regulators proactively uncover suspicious objects. In addition to the well-known abnormal objects that exist in the form of individuals, such as a rumor spreader in social media, some anomalies may exhibit collective behavior in reality, such as multiple fraudsters colluding with each other to garner benefits [1]. These anomalies and their interactions typically form subgraph structures. Effectively mining those unusual subgraphs in networks is of great significance to many high-impact domains [2], [3], [4]. For instance, as shown in Fig. 1, locating corporate groups that issue false and fictitious VAT invoices will help maintain a healthy and fair business environment. In a health surveillance network, the Centers for Disease Control and Prevention (CDC) staff have an interest in locating a collective of infected patients with significant abnormal symptoms, which may reveal a potential disease outbreak like COVID-19.

The superior performance of deep neural nets, especially graph neural networks [5], has promoted the rapid development of deep graph anomaly detection. However, most existing efforts [6], [7], [8] only focus on spotting individual graph anomalies, that is, single nodes or edges. In contrast, the abnormal subgraph (AS) detection task has received much less scrutiny despite the pressing need from real applications [1]. For instance, early methods [9], [10] exploit topological signals such as L_1 properties of the eigenvectors of the modularity matrix to reveal dense anomaly regions, yet the neglect of node attributes limits their identification of various intricate anomalies. Instead, AMEN [11] considers both attribute and topological information. It spots subgraph anomalies that exhibit special attribute distributions by measuring the

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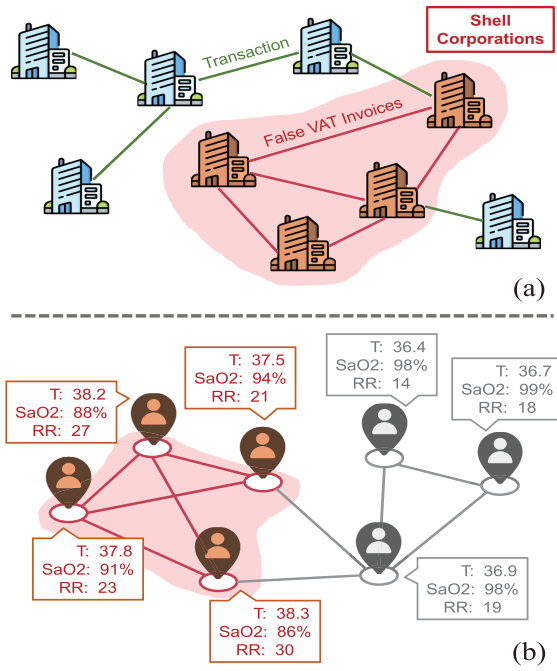


Fig. 1. (a) Fraudulent companies engage in tax evasion by issuing false VAT invoices to each other, often forming a subgraph with dense edge connections. (b) Group of patients infected with COVID-19 have unusual disease symptoms from healthy residents, that is, higher temperature (T), lower oxygen saturation (SaO₂), and higher respiratory rate (RR).

normality of the subgraph. However, the traditional data statistical technique used by AMEN results in low computational efficiency and poor scalability. Recent deep learning methods DeepFD [12] and FraudNE [13] first derive node embeddings through an autoencoder and then feed them into distribution density-based DBSCAN [14] to detect dense blocks as anomalies. However, this pipeline scheme is difficult to ensure that the learned representations are conducive to identifying anomalies and can easily lead to suboptimal solutions compared to end-to-end optimization. As a recent sophisticated end-to-end subgraph anomaly detection framework, AS-GAE [4] presents a location-aware graph autoencoder and a supermodular graph scoring function to uncover anomalous areas. However, its reliance on a manually set threshold is prone to mislocating abnormal areas, limiting its effectiveness in many complicated scenarios. Additionally, existing node anomaly detection approaches such as ANEMONE [15] and GRADATE [8] cannot be simply applied to identify subgraph-level anomalies as they lack recognition of subgraph structures and consideration of collaborative behaviors between nodes within the subgraph. More dedicated efforts are therefore needed to explore a promising solution for anomalous subgraph detection, which motivates our research work.

Combined with practical application scenarios, identifying subgraph anomalies faces the following challenging issues. First of all, it is inaccessible to acquire sufficient and high-quality ground-truth anomalies to support an end-to-end supervised learning paradigm. Due to the diversity of anomalies, in most cases, we have little or no prior knowledge about abnormal patterns in advance to label samples. Moreover, the combinatorial nature of subgraphs would make

the solution space exponentially large, which makes manual annotation almost impossible. Thus, exploring an unsupervised framework for AS detection is quite necessary. Second, deriving the anomaly-aware subgraph partition from both topological and attribute perspectives simultaneously can be a daunting task. Unlike independent nodes or edges, subgraphs composed of nodes and edges come in various sizes and inner structures due to their different construction forms. Meanwhile, the abnormality of subgraphs may be manifested in deviated attributes, dense topological connections, inconsistent associations between attributes and structures, and so forth. To reveal ASs, adaptively searching an anomaly-aware subgraph division through attribute and structural information is indispensable yet challenging, especially in unsupervised conditions where the potential subgraph space can be as high as exponential. Finally, it is challenging to design training objectives to effectively distinguish ASs from benign ones and quantify their degree of abnormality. The anomaly detection process necessitates appropriate objective functions to enable the detection model to capture the distinction between anomalies and benign counterparts. However, it is extremely hard to model the deviation or even quantify the arbitrary abnormal patterns within subgraphs when the ground-truth anomaly type is unknown. Although the previous nondeep methods [16], [17] resort to handcrafted measure functions, the poor expressive power of fixed features limits their ability to generalize to unseen anomalies.

To cope with the above challenges, we propose a end-to-end unsupervised subgraph anomaly detection framework (EndSubG), which jointly models subgraph partition and AS recognition in an end-to-end manner. Specifically, ENDSubG learns anomaly-aware graph embedding by explicitly capturing possible abnormal edges that violate the Homophily assumption [18]. Note that nodes exhibiting unusual collaborative behaviors have similar patterns and tend to form a subgraph. A trainable clustering layer is employed to learn subgraph partition according to semantic distribution while capturing the boundaries between anomalies and normal regions that contribute to uncovering anomalies. The learned soft clustering assignment ensures the possibility of nodes belonging to multiple subgraphs, which is more consistent with reality [19]. By coarsening the node and edge sets into subgraphs, we arrive at the subgraph-level composition of the input network. Then, a contrastive learning strategy that models the congruence between each subgraph and its vicinity derived from the coarsened subgraph network is introduced to measure the abnormality. Through end-to-end joint optimization, ENDSubG implements adaptive anomaly-aware graph embedding and subgraph division to facilitate the identification of ASs that deviate from the majority. In contrast, the two-stage pipeline method is prone to erroneously dividing the graph due to not directly targeting the anomaly detection task (e.g., the partitioning stage targets maximizing modularity [20] or graph reconstruction [21]), leading to failure in subgraph anomaly recognition. In practice, pipeline-based methods often involve the combination of various deep or nondeep models, resulting in an explosion of parameter combination space, from which selecting to obtain promising performance is very time-consuming. In addition, we design a new metric weighted normalized mutual information centered on AS (AS-

WNMI) that measures the quality of subgraph partitioning while assessing the ability to identify anomalies, which fills the gap in performance evaluation for the subgraph anomaly detection task. Our main contributions are summarized as follows.

- 1) We examine the limitations of existing works, then introduce a new idea that jointly models subgraph partition and subgraph anomaly detection as a whole instead of treating them as two separate steps.
- 2) We propose an end-to-end AS detection framework ENDSUBG based on anomaly-aware subgraph partition and subgraph-level contrastive learning, providing a new solution for the field of deep subgraph anomaly detection that has received less attention.
- 3) We design a new evaluation metric AS-WNMI to promote research on elaborate metrics for this task. Moreover, we empirically validate the superiority of ENDSUBG on both synthetic and real-world datasets, and intuitively explain and analyze the detected subgraphs through case studies.

II. RELATED WORKS

A. Node Anomaly Detection

Most existing efforts focus on identifying abnormal nodes following the unsupervised learning paradigm. Here, we briefly introduce the following three types of representative algorithms. Traditional shallow models such as Radar [22] and ANOMALOUS [23] resort to matrix factorization combined with residual analysis to spot abnormal nodes. With the rapid development of graph neural networks, deep frameworks such as DOMINANT [24], SpecAE [25], and AnomalyDAE [26] uncover anomalies utilizing GCN-based autoencoder with the capability of learning the expressive representations of nonlinear relationships within the complex structure. Subsequently, the powerful performance of contrastive learning brought new directions to the development of graph anomaly detection. Popular methods including CoLA [6], ANEMONE [15], and GRADATE [8] derive anomaly scores by learning to measure the agreement between contrastive counterparts. Recently, promising detection techniques have continued to be enriched. For instance, RG-GLD [27] achieves the lightweight anomaly detection for secure and efficient Internet-of-Things (IoT) communications by reconstructing the directed graph formed by data communications with global-local knowledge distillation. ARISE [28] spots topology anomalies and attribute anomalies simultaneously by perceiving suspicious substructures through a region proposal module. However, these methods aim to identify the abnormal behavior of a single entity rather than analyzing the discrepancy between node collaborative behaviors on a subgraph basis, making them unable to be directly applied in AS identification scenarios.

B. Subgraph Anomaly Detection

Unlike spotting node anomalies, anomaly detection on the subgraph level has received much less attention due to the difficulties in subgraph pattern modeling and subgraph abnormality quantification. One mainstream of existing works is nondeep techniques based on traditional graph analysis. For

instance, Miller et al. [9] provided a statistical model to detect less-correlated subgraphs through L1 properties of the graph's eigenvectors. SODA [16] designs a system that finds ASs given a graph and a query by indications of edge occurrence. Motivated by special attribute distributions of subgraph anomalies, gAnomaly [2], AMEN [11], and Slicendice [29] model attribute distributions and measure the normality of subgraphs for detection. Nonetheless, the dependency on hand-crafted anomaly measure functions makes it difficult for them to handle unknown anomaly patterns in the absence of prior information. Moreover, these methods cannot learn high-order semantic information from structure and attribute simultaneously, making them noise-sensitive and unable to generalize. Deep subgraph anomaly detection frameworks have recently shown superior performance and have become a hot research topic for future development. For instance, DeepFD [12] and FraudNE [13] project the user-item bipartite graph into node embeddings first and then find dense blocks through a density-based algorithm such as DBSCAN. However, the way that separates representation and anomaly detection as two independent steps easily leads to a suboptimal solution. In contrast, AS-GAE [4] develops an end-to-end detection framework. It employs a location-aware graph autoencoder to uncover anomalous areas based on reconstruction and a supermodular graph scoring function to measure the anomaly score of each subgraph. However, its reliance on a manually set threshold is prone to mislocating abnormal areas, and its inappropriate subgraph extraction strategy further leads to poor performance. To address the above limitations, we explore an end-to-end unsupervised subgraph anomaly detection method, which adaptively performs anomaly-aware graph embedding and subgraph extraction, then quantifies subgraph anomaly based on contrastive learning.

III. PROPOSED FRAMEWORK: ENDSUBG

Before going further, we first give a formal definition of the studied problem. Given an attributed network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ with $|\mathcal{V}| = n$ nodes and $|\mathcal{E}| = m$ edges, where each node $v_i \in \mathcal{V}$ is affiliated with a set of d -dimensional attributes (features) $\mathcal{F} = \{f_1, f_2, \dots, f_d\}$. We let the matrix $X \in \mathbb{R}^{n \times d}$ represent node attributes, where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the feature for node v_i . The adjacency matrix $A \in \{0, 1\}^{n \times n}$ is used to record edge links, where $A_{ij} = 1$ indicates the existence of an edge $e_{ij} = (v_i, v_j)$ between nodes v_i and v_j ; otherwise, $A_{ij} = 0$. A subgraph of $\mathcal{G}$ is denoted as $\mathcal{S} = (\mathcal{V}_\mathcal{S}, \mathcal{E}_\mathcal{S}, \mathcal{F})$, where $\mathcal{V}_\mathcal{S} \subseteq \mathcal{V}$ and $\mathcal{E}_\mathcal{S} \subseteq \mathcal{E} \cap \mathbb{P}(\mathcal{V}_\mathcal{S})$. $\mathbb{P}(\mathcal{V}_\mathcal{S})$ indicates the powerset of $\mathcal{V}_\mathcal{S}$. With the above notations, the goal of unsupervised AS detection aims to spot suspicious subgraphs $\mathcal{S} \subseteq \mathcal{G}$ that significantly deviate from the major patterns of the graph $\mathcal{G}$, where the likelihood of the subgraph being abnormal is quantified by a scoring function $f(\cdot)$. It should be emphasized that in this problem set, the candidate subgraph set $\tilde{\mathcal{S}} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_p\}$ for graph $\mathcal{G}$ is not given in advance. Considering that the potential subgraph search space is as large as exponential, our studied problem is more challenging than the setting where a candidate subgraph set is given beforehand like [3]. Next, we elaborate on the proposed end-to-end joint modeling framework ENDSUBG, a high-level overview of which is shown in Fig. 2.

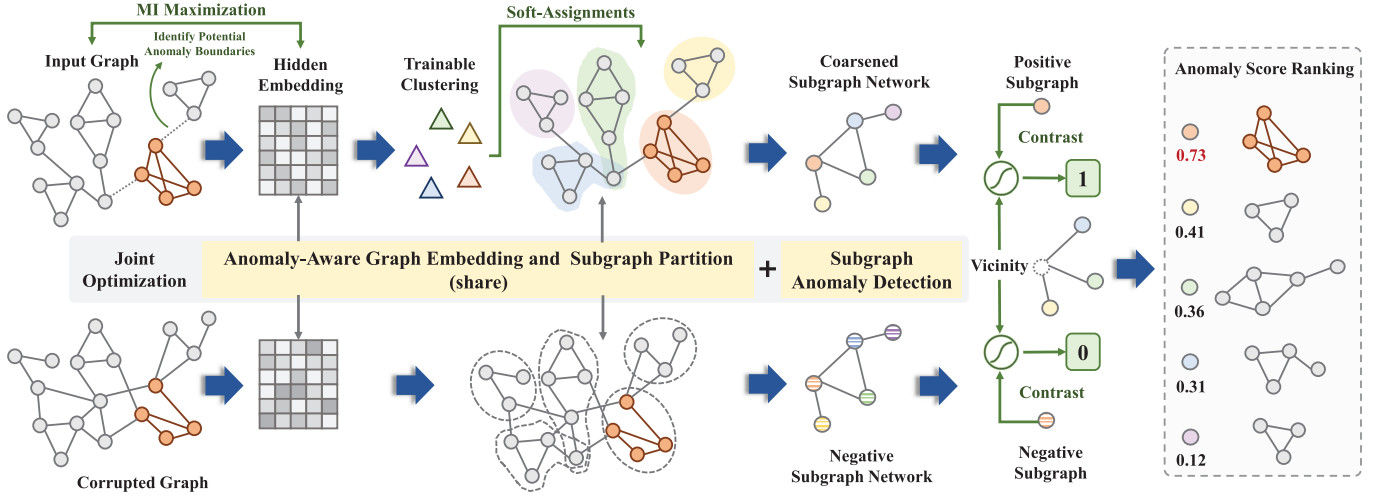


Fig. 2. High-level overview of the proposed ENDSUBG. Given an attributed graph, ENDSUBG achieves anomaly-aware graph embedding and subgraph partition by explicitly considering potential abnormal edges and capturing the boundaries between anomalies and normal regions. After forming the coarsened subgraph network, it spots ASs by learning the “subgraph-vicinity” matching patterns. ENDSUBG follows an idea that jointly models graph embedding, subgraph partition, and subgraph anomaly detection.

A. Anomaly-Aware Graph Embedding and Subgraph Partition

High-quality graph embedding is adept at capturing fine-grained pattern distinctions among nodes, which is essential for exposing unusual and problematic graph components to facilitate subgraph partition. Inspired by modern self-supervised graph representation learning [30], [31], we expect to build a graph embedding mechanism similar to GMI [32] that maximizes graphical mutual information between the input graph $\mathcal{G}$ and node embeddings $\mathbf{H} \in \mathbb{R}^{n \times d'}$. However, the existence of ASs may negatively impact representation quality through neighborhood message propagation and mutual information operations. Specifically, we find that the boundaries of ASs and normal regions often have illegal edge links that violate the Homophily principle. These edges have lower existence probabilities predicted by attribute similarity than others, introducing heterophily into the graph. However, many theoretical studies [33], [34], [35] have shown that conventional GNN models employ homophily as a strong inductive bias, and the heterophilic edges will seriously degrade their performance. Thus, capturing these boundaries helps GNN models perceive potential anomalies and focus on finely modeling the similarity of homophily signals. This facilitates good partitioning of subgraphs with similar collaborative behaviors. To this end, we refine the input graph topology by reweighting $\mathbf{A}$ with potential anomaly probabilities according to the well-known three-sigma rule (3σ rule) [36] in statistics. It states that almost all (99.73%) values are within three standard deviations of the mean, while values outside this range are most likely to be anomalies, which has been widely used in various fields for outlier detection [37], [38], [39]. Formally, for an edge e_{ij} , we estimate its anomaly probability $\mathbf{P}_{ij}$ by quantifying whether its existence probability is less than the average probability of all edges in $\mathcal{G}$ with a predefined number λ of deviation standards

$$\mathbf{P}_{ij} = \begin{cases} r_{ij}, & \text{if } r_{ij} < \text{MEAN}_{e_{kl} \in \mathcal{E}} r_{kl} - \lambda \text{STD}_{e_{kl} \in \mathcal{E}} r_{kl} \\ 1, & \text{otherwise} \end{cases} \quad (1)$$

where $r_{ij} = \cos(\mathbf{x}_i, \mathbf{x}_j)$ aims to calculate the existence probability of e_{ij} . Note that in practice, the standard deviation interval can be adjusted according to the requirements of different fields or tasks. To better adapt to distinct data distributions, we introduce the hyperparameter λ in (1) to control the threshold range. The experiments on parameter analysis also demonstrate the effectiveness of adjusting thresholds on different datasets. Then, we derive adjusted graph structure $\mathbf{A}' = \mathbf{A} \odot \mathbf{P}$, where $\odot$ denotes the Hadamard product.

Now, we perform anomaly-aware graph embedding by introducing weights $\mathbf{A}'$ to adjust the calculation of the graphical mutual information loss (see [32] for detailed theoretical analysis). We first define the computation graph of node v_i that specifies the neighborhood around a particular node to perform a localized convolution as $\mathcal{G}_i$, then maximize the mutual information between $\mathbf{h}_i \in \mathbb{R}^{d'}$ and $\mathcal{G}_i$ by

$$I(\mathbf{h}_i; \mathcal{G}_i) = I(\mathbf{h}_i; \mathbf{x}_i) + \sum_{j \in \mathcal{N}(i)} \frac{\mathbf{A}'_{ij}}{|\mathcal{N}(i)|} I(\mathbf{h}_i; \mathbf{x}_j) + I(w_{ij}; \mathbf{A}'_{ij}) \quad (2)$$

where $w_{ij} = \sigma(\mathbf{h}_i^\top \mathbf{h}_j)$ with the sigmoid function σ . Under the guidance of weight $\mathbf{A}'_{ij}$, the representation $\mathbf{h}_i$ will extract and preserve less information about neighboring nodes that may be anomalies while weakening the topological tie with the possibly abnormal nodes to form a boundary. As for $I(\mathbf{h}_i; \mathbf{x}_j)$, we can employ well-known mutual information estimators such as the Jensen-Shannon MI estimator (JSD) [40] to calculate (see experiments for the other estimator)

$$I(\mathbf{h}_i; \mathbf{x}_j) = -sp(-\mathcal{D}_\theta(\mathbf{h}_i, \mathbf{x}_j)) - \mathbb{E}_{v_n \sim \mathbb{P}_n} [sp(\mathcal{D}_\theta(\mathbf{h}_i, \mathbf{x}_{v_n}))] \quad (3)$$

where $\mathbb{P}_n$ denotes the uniform distribution on the node set $\mathcal{V} \setminus \{v_j\}$, and $sp(x) = \log(1 + e^x)$ is the soft-plus function. The discriminator $\mathcal{D}_\theta : d' \times d \rightarrow \mathbb{R}$ we applied is a simple bilinear scoring function

$$\mathcal{D}_\theta(\mathbf{h}_i, \mathbf{x}_j) = \sigma(\mathbf{h}_i^\top \theta \mathbf{x}_j) \quad (4)$$

where θ denotes a learnable parameter and the logistic sigmoid nonlinearity σ aims to convert scores into probabilities of $(\mathbf{h}_i, \mathbf{x}_j)$ being a positive pair. In addition, we maximize $I(w_{ij}; \mathbf{A}'_{ij})$ via calculating an adjusted cross-entropy

$$I(w_{ij}; \mathbf{A}'_{ij}) = \mathbf{A}_{ij} \log \frac{w_{ij}}{\mathbf{A}'_{ij}} + (1 - \mathbf{A}_{ij}) \log(1 - w_{ij}). \quad (5)$$

Now, we arrive at the objective loss for graph embedding

$$\mathcal{L}_{enc} = - \sum_{v_i \in \mathcal{V}} I(\mathbf{h}_i; \mathcal{G}_i). \quad (6)$$

By maximizing $I(\mathbf{h}_i; \mathcal{G}_i)$ with the sum of (3) and (5) over all hidden vectors $\mathbf{H}$, the learned representations finely preserve the attribute and topological information inherent in the graph, and has good separability between benign and abnormal, which helps in adaptive division of subgraphs.

Considering that nodes exhibiting collaborative behaviors typically have similar patterns, leading to their high proximity in the embedding space. This inspires us to utilize a differentiable clustering layer [41] for adaptive subgraph partitioning based on the semantic distribution of representations. As the model trains, this layer continuously optimizes the soft assignment of nodes to clusters (i.e., subgraphs), then the above nodes will tend to be grouped into a subgraph.

Let μ_c indicate the center of cluster $c \in \{1, 2, \dots, C\}$. The assignment matrix to be learned is denoted as $\mathbf{S} \in \mathbb{R}^{n \times C}$, where S_{ic} records the degree to which node v_i is assigned to cluster c . Here, we utilize a soft-min assignment to assign each node v_i to the cluster centers based on distance

$$S_{ic} = \frac{\exp(-\tau \|\mathbf{h}_i - \mu_c\|)}{\sum_C \exp(-\tau \|\mathbf{h}_i - \mu_c\|)} \quad (7)$$

where τ is an inverse-temperature hyperparameter, and we use norm $\|\cdot\|$ as negative cosine similarity due to its good performance. The soft assignments $\mathbf{S}$ are more adaptable to the assumption that nodes may participate in multiple clusters simultaneously. Similar to typical K -means, we optimize each cluster center by alternately updating S_{ic} via (7) and $\mu_c = (\sum_i S_{ic} \mathbf{h}_i / \sum_i S_{ic})$. This iterative process will converge to a fixed point where μ remains unchanged between successive updates [42].

Intuitively, the more common subgraphs node v_i and node v_j participate in, the more likely they are to be connected by an edge. At the same time, we also need to force nodes at the boundary to be divided into different clusters. Hence, we aim to apply $\mathbf{A}'_{ij}$ carrying anomaly boundary information to constrain the similarity and dissimilarity between cluster assignment vectors of different nodes

$$\mathcal{L}_{par} = - \sum_{(v_i, v_j) \in \mathcal{E}} \log \left(\frac{\sigma(S_i S_j^T)}{\mathbf{A}'_{ij}} \right) - \mathbb{E}_{\mathbb{P}_n} [\log(\sigma(-S_n S_m^T))] \quad (8)$$

where $\mathbb{P}_n$ denotes the uniform distribution over nonedges (v_n, v_m) in $\mathcal{G}$. As can be seen, the trainable clustering layer will adaptively group nodes with similar patterns that may belong to normal or abnormal subgraphs. Based on the subgraph composition of graph $\mathcal{G}$, we switch the perspective from node level to subgraph level to examine the abnormality of each subgraph, which is the focus in Section III-B.

It should also be noted that the behavior of subgraph anomaly detection heavily depends on the quality of extracted subgraphs, especially in unsupervised scenarios. Existing methods have paid little attention to this aspect and still have many limitations. For instance, AMEN treats the ego-net of each node as a subgraph to be examined. Although it reaches fine granularity, it ignores the role of attribute correlation in subgraph division. AS-GAE extracts subgraphs by finding connected components in the residual network. This partitioning fails to capture the cooperativity exhibited by subgraph anomalies, leading to undesirable results. In contrast, the learnable clustering strategy we adopt achieves adaptive anomaly-aware subgraph partition, which is beneficial to the subsequent accurate identification of subgraph anomalies.

B. Subgraph-Level Contrastive Learning

For the exported candidate subgraph set $\{S_1, S_2, \dots, S_C\}$, the key question is how to measure the abnormality of each subgraph in an unsupervised manner to distinguish anomalies from benign. Note that ASs often exhibit discrepancies with their surrounding areas, serving as a good anomaly indicator. Hence, we focus on picking out this inconsistency by learning the “subgraph-vicinity” matching patterns. In specific, we first convert the input graph $\mathcal{G}$ into a subgraph-level composition form through the following coarsening operations:

$$\begin{aligned} \mathbf{X}_S &= \mathbf{S}^T \mathbf{H} \in \mathbb{R}^{C \times d'} \\ \mathbf{A}_S &= \mathbf{S}^T \mathbf{A} \mathbf{S} \in \mathbb{R}^{C \times C} \end{aligned} \quad (9)$$

where $\mathbf{X}_S$ stores hidden embeddings for each of the subgraphs, which is derived by aggregating node embeddings according to the learned cluster assignments. Similarly, $\mathbf{A}_S$ is the coarsened subgraph adjacency matrix that records the connectivity strength between each pair of subgraphs.

To adapt to the lack of prior annotations, we create “subgraph versus vicinity” instance pairs and model their matching associations in a contrastive learning manner. For each subgraph S_i , its vicinity $\mathbf{V}_i$ is defined by the fused neighboring subgraphs $\mathbf{V}_i = \sum_{j \in \mathcal{N}(i)} \mathbf{A}_{S_{ij}} \mathbf{X}_{S_j}$, which together form a positive pair. To generate negative instances, we build a corrupt graph $\tilde{\mathcal{G}} = (\tilde{\mathbf{X}}, \mathbf{A})$ by adding noise sampled from a uniform distribution to the given attributes, that is, $\tilde{\mathbf{X}} = \mathbf{X} + \epsilon$. After passing through the same graph encoding layer, we get negative subgraph representation $\tilde{\mathbf{X}}_S$ by $\mathbf{S}^T \tilde{\mathbf{H}}$. The objective function follows a standard binary cross-entropy loss to make the discriminator successfully differentiate between positive and negative examples:

$$\mathcal{L}_{con} = -\frac{1}{2C} \sum_{i=1}^C (\log \mathcal{D}_{\Psi}(\mathbf{X}_{S_i}, \mathbf{V}_i) + \log(1 - \mathcal{D}_{\Psi}(\tilde{\mathbf{X}}_{S_i}, \mathbf{V}_i))) \quad (10)$$

where $\mathcal{D}_{\Psi}: d' \times d' \rightarrow \mathbb{R}$ is a discriminator similar to (4) with trainable parameters Ψ .

C. Joint Optimization for Subgraph Anomaly Detection

By combining the above three objectives through tradeoff parameters α and β , we get the complete objective function for ENDSUBG

$$\mathcal{L} = \alpha \mathcal{L}_{enc} + \beta \mathcal{L}_{par} + \mathcal{L}_{con}. \quad (11)$$

After end-to-end joint training of graph embedding, subgraph partition, and anomaly identification, we acquire a useful classifier $\mathcal{D}_\Psi$ to discriminate the consistency between a subgraph and its vicinity. Then, the abnormality of each subgraph $\mathbf{a}_i$ can be quantified by the scoring function $f = 1 - \mathcal{D}_\Psi(\mathbf{X}_{S_i}, \mathbf{V}_i)$, where $\mathbf{a} \in \mathbb{R}^C$ records the anomaly scores of all subgraphs. Multiple rounds of statistical analysis can also be used to calculate anomaly scores [6], [15]. Due to the mismatching with surrounding areas, an AS will be assigned a lower probability of being positive (close to 0) by $\mathcal{D}_\Psi$, thus having a higher anomaly score. Interestingly, we can then evaluate the abnormality of each node through $\mathbf{S} \cdot \mathbf{a}$, where the anomalous nodes will provide node-level explanations for the subgraph anomaly. In other words, for the detected AS, we can further figure out the abnormality of each node, which is of great significance for subsequent analysis, management, and even rectification of anomalies.

D. Complexity Analysis

The graph embedding stage adopts a vanilla GCN [43] encoder, which has the computational complexity of $O(lndd' + lmd')$, where l denotes the number of layers. The discriminator $\mathcal{D}_\Theta$ has complexity $O(ndd' + nd)$, and the clustering assignment needs $O(nd'C)$. During the network coarsening phase, it needs $O(nd'C + mC + C^2n)$. The embedding of corrupt graph also costs $O(lndd' + lmd')$, and the discriminator $\mathcal{D}_\Psi$ has complexity $O(Cd'^2 + Cd')$. Note that l and C are usually much smaller than n , m , d , and d' , the overall time complexity is $\text{num_iters} * O(ndd' + md')$, which is linearly related to the number of nodes and edges.

IV. EXPERIMENTS

In this section, we propose a new metric specifically designed to evaluate the quality of subgraph anomaly detection, which fills the gap in performance evaluation for this task. Then, we empirically evaluate ENDSUBG on both synthetic and real-world datasets. A case study is also given to intuitively demonstrate the quality of ASs identified by ENDSUBG and other baselines. Specifically, we mainly attempt to answer the following four research questions.

- 1) *Q1*: How to quantitatively evaluate the performance of an algorithm in identifying ASs?
- 2) *Q2*: How impactful is the proposed framework on synthetic datasets with different background graphs?
- 3) *Q3*: Does the proposed framework work well on real-world datasets with different numbers of ASs?
- 4) *Q4*: How to understand ASs identified by the proposed framework (visual analysis)?

A. New Evaluation Metric: AS-WNMI (Answer to *Q1*)

Currently, there is no unified specification on how to evaluate the performance of AS detection. Most existing works [4], [13] still employ commonly used area under the curve (AUC) or $F1$ to assess detection results at the node level, which completely ignores the role of subgraphs. Besides, it should be noted that in the task setting of unsupervised subgraph anomaly detection, there is no pre-given candidate subgraph set $\bar{\mathcal{S}} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_C\}$ with subgraph anomaly labels. Instead of

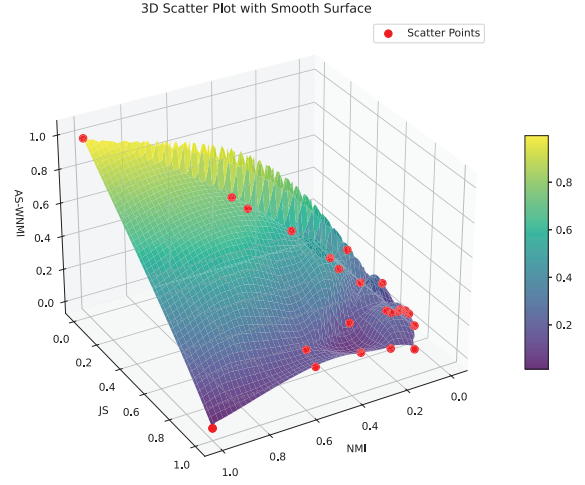


Fig. 3. 3-D surface plot of AS-WNMI, NMI, and JS distance values under different simulation cases.

simply judging whether each subgraph $\mathcal{S}_i$ in the given set is abnormal, each method utilizes different partition strategies as discussed in Section III-A to get subgraphs of various shapes and then spots anomalies accordingly. In this case, metrics similar to classification accuracy cannot be used. To meet different application needs, more dedicated research on new metrics that evaluate anomaly detection quality from the subgraph level is urgently needed [1].

It can be observed that the quality of the identified ASs depends on the correctness of the extracted subgraphs and abnormality measurements. Accordingly, we introduce the anomaly prediction difference as a weight into NMI (often used to measure the quality of community detection [19]) and propose an AS-WNMI as a new evaluation metric. Assume that the ground-truth subgraph anomaly labels consist of the subgraph set $\bar{\mathcal{S}}^*$ and node anomaly labels $\mathbf{y}^* = [y_1^*, y_2^*, \dots, y_n^*] \in \{0, 1\}^n$, where all nodes belonging to ASs $\mathcal{S}_i^*$ are labeled with 1, and others are marked as 0. Similarly, the outputs of the detection model include the predicted subgraph set $\bar{\mathcal{S}}$ and node anomaly scores $\mathbf{y} = [y_1, y_2, \dots, y_n] \in [0, 1]^n$. First, unlike the community detection task that focuses on whether each community (subgraph) partition is correct, our task concentrates on extracting ASs rather than the division of normal regions. That is, normal regions can be divided in any way that helps identify ASs. Thus, we prune $\bar{\mathcal{S}}^*$ by merging all normal subgraphs into one to omit concerns about the partitioning results of normal regions. And do the same for $\bar{\mathcal{S}}$. (If $\bar{\mathcal{S}}^*$ or $\bar{\mathcal{S}}$ only involves ASs without further subgraph division of normal regions, the above operation is unnecessary.) Then, the AS-WNMI ($\bar{\mathcal{S}}^*, \bar{\mathcal{S}}, \mathbf{y}^*, \mathbf{y}$) score is defined as

$$\frac{-2 \sum_{i=1}^{C^*} \sum_{j=1}^C n_{ij} \log \frac{n_{ij}n}{n_i n_j} \left(1 - \text{JS}(\mathbf{y}_{\mathcal{S}_i^*}^*, \mathbf{y}_{\mathcal{S}_j}\right)}{\sum_{i=1}^{C^*} n_i \log \frac{n_i}{n} + \sum_{j=1}^C n_j \log \frac{n_j}{n}} \quad (12)$$

where C^* and C are the number of ground truth and detected subgraphs, respectively. n_{ij} denotes the number of nodes appearing in both the i th ground-truth subgraph $\mathcal{S}_i^*$ and the j th detected subgraph $\mathcal{S}_j$. n is the number of nodes in $\mathcal{G}$. n_i and n_j sum up the number of nodes in $\mathcal{S}_i^*$ and $\mathcal{S}_j$, respectively. $\mathbf{y}_{\mathcal{S}_i^*}^*$ denotes the anomaly labels of nodes belonging to $\mathcal{S}_i^*$, and

y_{S_j} is the predicted anomaly scores of nodes belonging to S_j . $JS(\mathbf{p}, \mathbf{q})$ calculates the Jensen–Shannon distance between two probability vectors $\mathbf{p}$ and $\mathbf{q}$, which is the square root of the Jensen–Shannon divergence and defined as

$$\sqrt{\frac{KL(\mathbf{p} \parallel \mathbf{m}) + KL(\mathbf{q} \parallel \mathbf{m})}{2}} \quad (13)$$

where $KL(\cdot)$ denotes the Kullback–Leibler divergence, and $\mathbf{m} = (\mathbf{p} + \mathbf{q})/2$. By measuring the anomaly prediction difference between $y_{S_i}^*$ and y_{S_j} (the smaller the difference, the value of $JS(y_{S_i}^*, y_{S_j})$ approaches 0, otherwise, it approaches 1), (13) quantifies the anomaly recognition performance of the detection model on the subgraph from the node level. After introducing $[1 - JS(y_{S_i}^*, y_{S_j})]$ as weight into NMI, (12) measures the quality of subgraph partitioning and simultaneously evaluates the ability to identify anomalies in subgraphs. Here, a simulation experiment is used to illustrate the superiority of the proposed AS-WNMI over the vanilla NMI. Let us consider a simple situation. There exist only two clusters, the first one consisting of five nodes is labeled as normal cluster and the second one with three nodes is abnormal. We enumerate all possible partitioning cases ($2^8 = 256$) and all possible anomaly cluster prediction results ($2! = 2$), leading to $256 \times 2 = 512$ possible outcomes. We then calculate the scores for each of the three metrics (i.e., NMI, JS distance, and AS-WNMI) in each case and draw them in a 3-D surface plot, as shown in Fig. 3. As can be seen, the value range of AS-WNMI belongs to $[0, 1]$ and will reach its maximum value of 1 only when NMI reaches 1 and JS distance reaches 0. In other words, AS-WNMI can reach its highest value only when the AS is correctly divided from the normal region and all nodes in each part are correctly judged, that is, the nodes in the AS are predicted to be anomalous and the nodes in the normal region are determined as benign. In this sense, the evaluation perspective of the AS-WNMI metric is more comprehensive compared with NMI which only measures the quality of subgraph partition, and the traditional AUC or average precision (AP) metric which only considers whether abnormal nodes are identified. Thus, obtaining a high AS-WNMI score is somewhat challenging for the detection model.

B. Compared Methods

It needs to be emphasized again that, there are limited existing studies on unsupervised static subgraph anomaly detection, as mentioned in the related work. Thus, we select the following four categories of methods as baselines to meet the focus of our work.

1) *Node Anomaly Detection*: Two state-of-the-art methods are selected to explore whether they are still effective in subgraph anomaly detection, which are both based on self-supervised contrastive learning.

- 1) *ANEMONE* [15]: spots anomalous nodes by learning the patch-level (i.e., node versus node) agreement and context-level (i.e., node versus ego-net) agreement concurrently.
- 2) *GRADATE* [8]: presents a multiscale contrastive learning framework with an augmented graph view for node anomaly detection.

2) *Dense Block Detection*: Following the idea of DeepFD [12] and FraudNE [13] that learn bipartite graph embedding first and then uncover suspicious dense blocks with DBSCAN [14] for fraud detection, we design a baseline consisting of two phases: attributed graph representation learning and dense block detection.

- 1) *GAE* [44] + *DBSCAN*: GAE generates node embeddings by learning to reconstruct topological links, after which DBSCAN identifies dense regions in the vector space.
- 2) *DGI* [30] + *DBSCAN*: DGI derives informative node embeddings via maximizing the mutual information between patch-level and graph-level representations and then achieves dense block detection by DBSCAN.
- 3) *Subgraph Anomaly Detection (Pipeline)*: For an arbitrary subgraph set, SADE [3] learns role-guided subgraph embedding first by taking into consideration both local connections and global roles and then employs classic unsupervised outlier detection techniques such as Isolation Forest [45] or LOF [46] to spot suspicious transactions in the encoded subgraph embedding space. Since there is no predefined subgraph set under our task setting, we adapt SADE to a baseline that follows a pipeline of subgraph division, subgraph embedding, and outlier detection.

- 1) *Louvain* [47] + *SADE*: Louvain divides the input graph into multiple communities based on modularity optimization, which is fed into SADE to derive subgraph embeddings, and then LOF spots outliers in the vector space by measuring the local deviation of the density of a certain object with respect to its neighbors.
- 2) *CommDGI* [20] + *SADE*: CommDGI performs clustering assignment by learning to maximize community mutual information and modularity from both attribute and topological perspectives and then identifies anomalies in the subgraph embedding space with the help of SADE.
- 4) *Subgraph Anomaly Detection (End-to-End)*: We compare ENDSUBG with existing two open-source end-to-end methods, including a traditional nondeep model and a deep learning framework.

- 1) *AMEN* [11]: introduces a measure called normality to evaluate the abnormality of attributed neighborhoods from both internal consistency and external separability.
- 2) *AS-GAE* [4]: learns a residual graph by location-aware graph autoencoder, on which a trainable supermodular graph scoring function is applied to assign anomaly scores to the extracted subgraphs.

C. Parameter Settings and Metrics

All experiments, except AMEN based on MATLAB, are implemented in PyTorch and carried out on a single NVIDIA 4090 GPU. Since SADE does not have a public source code, we implemented it ourselves based on its published paper. As for other baselines, they are all implemented using the released source code and undergo parameter tuning according to the empirical strategies reported in their papers. About ENDSUBG, we adopt two layers of GCN with 512 or 128 neurons as the graph encoder. The number of negative samples for the JSD estimator is set to 5. We select the number of

TABLE I
STATISTICS OF SYNTHETIC AND REAL-WORLD DATASETS

		#Nodes	#Edges	#Attributes	#Abnormal Subgraph	#Anomalous Nodes	Anomaly Ratio
Synthetic	ER Graph	4,880	137,600	128	4	286	5.86%
	BA Graph	4,717	42,787	128	4	274	5.81%
	CWS Graph	4,683	135,208	128	4	286	6.11%
Real-World	Reddit_1AS	10,618	90,492	64	1	665	6.26%
	Reddit_3AS	10,618	88,547	64	3	786	7.40%
	Questions_1AS	47,461	183,962	301	1	2,753	5.80%
	Questions_3AS	47,461	178,586	301	3	2,878	6.06%
	LargeCora_1AS	11,471	35,245	3780	1	646	5.63%
	LargeCora_3AS	11,471	34,324	3780	3	784	6.83%
	AmazonComputer_1AS	13,471	259,407	767	1	788	5.85%
	AmazonComputer_3AS	13,471	252,424	767	3	959	7.12%

extracted subgraphs C from $\{8, 10, 12, 15, 20\}$, and the initial cluster centers are chosen by the idea of K -means++, which benefits to accelerate the convergence of the model [48]. During training, we adopt the Adam [49] optimizer with the learning rate adjusted in the range of $\{1e^{-3}, 3e^{-4}, 1e^{-4}\}$. The inverse-temperature hyperparameter τ is set to 30. The parameters α and β are selected from $\{0.1, 0.3, 0.5, 0.7, 1.0\}$, and we tune the parameter λ in $[1, 2]$ combined with the interquartile range (IQR) rule [50].

On the one hand, following the commonly used metrics in the previous subgraph anomaly detection works [4], [11], which only consider whether node anomalies are spotted, we also calculate AUC and AP based on y^* and y to assess the detection results. The AUC score (area under the receiver operating characteristic curve, ROC-AUC) assesses how well the model generates relative scores to distinguish positive or negative examples across all classification thresholds. The AP score evaluates the ability of the model to identify positive instances accurately while maintaining high precision. AP is more suitable for anomaly detection tasks where positive (abnormal) and negative (benign) samples are extremely imbalanced. On the other hand, we adopt our proposed AS-WNMI to comprehensively evaluate the subgraph anomaly detection performance from both subgraph partitioning and anomaly identification. Here, we report the average results over five runs for the above metrics.

D. Results on Synthetic Datasets (Answer to Q2)

1) *Data Generation*: To simulate different scenarios comprehensively, we consider the following three network models as background graphs: 1) Erdős-Rényi (ER) model that generates a graph where a given number of nodes are randomly connected with a certain probability; 2) Barabási-Albert (BA) model that creates scale-free graphs; 3) connected Watts-Strogatz (CWS) model that produces connected small-world property graphs. We choose one specific type above and utilize the following NetworkX graph generators to generate 20 independent subgraphs with different numbers of nodes and structural properties sampled from a given distribution range.

- 1) ER graph is generated with the function `erdos_renyi_graph(n, p)`, where n for the number of nodes and p for the probability for edge creation.

TABLE II
PARAMETER SETTINGS FOR NETWORK APIS

		Normal Subgraph	Anomalous Subgraph
ER	n	<code>rint(180, 360)</code>	<code>rint(60, 80)</code>
	p	<code>uni(0.10, 0.15)</code>	<code>uni(0.25, 0.35)</code>
BA	n	<code>rint(180, 360)</code>	<code>rint(60, 80)</code>
	m	<code>rint(2, 4)</code>	<code>rint(5, 8)</code>
	p	<code>uni(0.25, 0.37)</code>	<code>uni(0.18, 0.29)</code>
	q	<code>uni(0.18, 0.29)</code>	<code>uni(0.25, 0.37)</code>
CWS	n	<code>rint(180, 360)</code>	<code>rint(60, 80)</code>
	k	<code>uni(0.12, 0.15) * n</code>	<code>uni(0.23, 0.28) * n</code>
	p	<code>uni(0.25, 0.37)</code>	<code>uni(0.18, 0.29)</code>

- 2) BA graph is created with the function `extended_barabasi_albert_graph(n, m, p, q)`, where n for the number of nodes, m for the number of edges with which a new node attaches to existing nodes, p for the probability value for adding an edge between existing nodes, and q for the probability value of rewiring of existing edges.
- 3) `connected_watts_strogatz_graph(n, k, p)` is for the CWS graph, where n for the number of nodes, k for the number of nearest neighbors in a ring topology, and p for the probability of rewiring each edge.

We select four from all subgraphs and label them as anomalies by setting parameters to make them smaller in network size and denser in topological structure. Then, we connect all subgraphs together to form a connected graph by setting a probability of 0.3 of one node connecting to any node in the other subgraph. In addition, node feature design follows the pattern provided by AS-GAE [4]. Node features in normal subgraphs and ASs are sampled from different Poisson distributions with randomly selected λ and the constraint $(\lambda_{\text{anomaly}}/\lambda_{\text{normal}}) = 5$. The definition of a Poisson distribution is $P_{\lambda}(x) = (\lambda^x e^{-\lambda}/x!)$. The data statistics are summarized in Table I. All parameters are randomly selected from a specific distribution shown in Table II.

2) *Performance Evaluation*: Table III reports AUC, AP, and AS-WNMI scores of all methods, based on which we have the following observations.

- 1) Across all datasets, ENDSUBG achieves almost the best results concerning all metrics, except AS-WNMI on the BA graph (detailed explanations of this special case later). The best AUC and AP scores reflect the superiority of ENDSUBG in discovering anomalies

TABLE III

PERFORMANCE OF DIFFERENT METHODS ON SYNTHETIC DATASETS. THE BEST RESULTS ARE MARKED IN BOLD. N/A MEANS NOT APPLICABLE

Metric	AUC	BA Graph AP	AS-WNMI	AUC	CWS Graph AP	AS-WNMI	AUC	ER Graph AP	AS-WNMI
ANEMONE	0.314	0.039	N/A	0.843	0.284	N/A	0.736	0.203	N/A
GRADATE	0.504	0.063	N/A	0.886	0.299	N/A	0.846	0.257	N/A
GAE+DBSCAN	0.968	0.881	0.735	0.607	0.076	0.028	0.593	0.071	0.031
DGI+DBSCAN	0.959	0.920	0.748	0.536	0.130	0.061	0.526	0.107	0.055
Louvain+SADE	0.519	0.162	0.146	0.463	0.068	0.021	0.452	0.082	0.089
CommDGI+SADE	0.861	0.401	0.108	0.759	0.320	0.103	0.549	0.068	0.052
AMEN	0.660	0.112	0.078	0.944	0.519	0.109	0.934	0.322	0.178
AS-GAE	0.924	0.290	0.251	0.925	0.305	0.214	0.924	0.293	0.253
ENDSUBG (ours)	0.993	0.977	0.376	0.995	0.986	0.384	0.998	0.997	0.343

from the node granularity compared to other baselines. Besides, the promising AS-WNMI value shows that ENDSUBG works well in spotting subgraph anomalies from the perspective of subgraph division and anomaly identification in an unsupervised fashion. This advantage benefits from the fact that the anomaly-aware graph embedding and subgraph partition explicitly capture the boundary between anomaly and benign, which facilitates the discovery of subgraph anomalies through subgraph-neighborhood contrastive learning. Moreover, our ENDSUBG is relatively robust to different background graph types.

- 2) Overall, dense block detection models and pipeline-based methods exhibit undesirable performance on this task, although GAE + DBSCAN and DGI + DBSCAN get good results in the BA graph. Note that the BA graph is a scale-free network that is closer to reality and much sparser than the other two graphs (see Table III). ASs are obviously denser in the network and are therefore easily recognized by dense block detection methods. However, in the denser CWS graph and ER graph, it is difficult to distinguish the boundary between ASs and normal regions based on density alone, so the performance is depressing. Moreover, the high AS-WNMI score on the BA graph is since the DBSCAN generates clear anomaly and benign labels rather than anomaly scores (probabilities), that is, the prediction $\mathbf{y} = [0, 1, \dots, 1] \in \{0, 1\}^n$ instead of $\mathbf{y} \in [0, 1]^n$, which will result in a higher AS-WNMI value due to less error when calculating $JS(\mathbf{y}_{S_i}^*, \mathbf{y}_{S_j})$. Methods that output anomaly probabilities suffer slightly in this regard, but the anomaly warning level supported by the probability is more suitable for practical application. The higher AUC and AP scores still reflect the stronger anomaly identification ability of ENDSUBG. Moreover, the large parameter search space caused by the combination of multiple methods leads to a formidable defect. It should also be noted that the synthetic datasets here simulate an ideal detection scenario with obvious subgraph anomalies. While these methods are somewhat effective, whether they still work well in more complicated real-world situations will be explored later.
- 3) Although two node anomaly detection models still uncover a certain portion of anomalies, due to the lack of recognition of the subgraph structure, their detection

outputs do not form the concept of subgraph anomalies, thus inapplicable to AS-WNMI. Besides, the sparseness of the BA graph makes it difficult to ensure the diversity of sampled contrast pairs based on random walk with restart (RWR), thus impairing the correct identification of anomalies by the discriminator.

E. Results on Real-World Datasets (Answer to Q3)

1) *Data Description and Anomaly Injection:* Note that there are currently almost no real-world datasets for AS detection publicly available in the research community. Privacy policy aside, manually spotting and annotating anomalous subgraphs is inherently challenging. Thus, we select four real datasets here, then following the previous studies [4], [51] inject ASs for experimentation.

LargeCora is an extension of the common citation network Cora [43], where nodes correspond to documents and edges indicate citation relations. Each node is associated with a bag-of-words representation whose dimension is determined by the dictionary size. AmazonComputer is the “Computer” part of the Amazon co-purchase graph [52], where nodes represent computers, edges indicate that two products are frequently bought together, and node features are bag-of-words encoded product reviews. Reddit [53] contains a user-subreddit graph that captures posts shared across subreddits for a month. The nodes represent users, whose interactions with posts form edges, and posts are converted into node features. Questions [54] is collected from the question-answering website Yandex Q, where nodes are users and edges indicate that one user answered the other user’s question during one year. Each user has a description that is converted into an average of FastText embeddings for words. Since the latter two datasets contain annotated node anomalies, they need to be removed before injecting anomalous subgraphs.

Following the method outlined in [51], we inject one or three ASs into each dataset to simulate two detection scenarios. Taking the injection of one anomaly as an example, we first find two connected subgraphs from the graph randomly, one of which is larger and the other is smaller. Then, we select a connected subset of nodes in the larger subgraph that matches the size of the smaller subgraph as a candidate for subgraph anomaly. The node attributes of this subset are replaced by those of the smaller subgraph to ensure a collective attribute distribution. Moreover, edge links are randomly

TABLE IV
PERFORMANCE OF DIFFERENT METHODS ON REAL-WORLD DATASETS WITH ONE AS. THE BEST RESULTS ARE MARKED IN BOLD

Metric	LargeCora_1AS			Reddit_1AS			AmazonComputer_1AS			Questions_1AS		
	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI
ANEMONE	0.934	0.491	N/A	0.670	0.112	N/A	0.889	0.267	N/A	0.656	0.089	N/A
GRADATE	0.906	0.361	N/A	0.663	0.082	N/A	0.811	0.182	N/A	0.708	0.108	N/A
GAE+DBSCAN	0.520	0.073	0.013	0.614	0.079	0.033	0.553	0.065	0.023	0.513	0.066	0.009
DGI+DBSCAN	0.502	0.071	0.002	0.590	0.075	0.034	0.504	0.063	0.008	0.513	0.066	0.009
Louvain+SADE	0.923	0.505	0.209	0.937	0.407	0.144	0.761	0.165	0.115	0.435	0.103	0.083
CommDGI+SADE	0.667	0.131	0.074	0.896	0.260	0.150	0.717	0.126	0.068	0.609	0.096	0.051
AMEN	0.420	0.061	0.048	0.785	0.212	0.140	0.707	0.179	0.126	0.432	0.056	0.042
AS-GAE	0.473	0.071	0.004	0.535	0.070	0.006	0.543	0.068	0.008	0.540	0.079	0.007
ENDSUBG _{JSD}	0.963	0.870	0.264	0.992	0.954	0.274	0.950	0.387	0.138	0.859	0.342	0.106
ENDSUBG _{InfoNCE}	0.958	0.862	0.259	0.994	0.957	0.283	0.960	0.388	0.141	0.869	0.357	0.112

TABLE V
PERFORMANCE OF DIFFERENT METHODS ON REAL-WORLD DATASETS WITH THREE ASS. THE BEST RESULTS ARE MARKED IN BOLD

Metric	LargeCora_3AS			Reddit_3AS			AmazonComputer_3AS			Questions_3AS		
	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI
ANEMONE	0.814	0.207	N/A	0.734	0.204	N/A	0.745	0.190	N/A	0.593	0.069	N/A
GRADATE	0.753	0.164	N/A	0.697	0.155	N/A	0.797	0.268	N/A	0.615	0.089	N/A
GAE+DBSCAN	0.582	0.080	0.011	0.623	0.095	0.043	0.541	0.077	0.019	0.512	0.055	0.006
DGI+DBSCAN	0.501	0.063	0.001	0.588	0.088	0.035	0.504	0.074	0.008	0.513	0.055	0.006
Louvain+SADE	0.518	0.088	0.082	0.654	0.123	0.115	0.565	0.119	0.105	0.629	0.124	0.104
CommDGI+SADE	0.655	0.114	0.067	0.633	0.167	0.138	0.499	0.071	0.010	0.515	0.056	0.003
AMEN	0.433	0.056	0.057	0.585	0.129	0.120	0.562	0.117	0.113	0.440	0.048	0.047
AS-GAE	0.479	0.064	0.002	0.512	0.077	0.003	0.558	0.014	0.014	0.560	0.079	0.014
ENDSUBG _{JSD}	0.853	0.335	0.188	0.859	0.589	0.221	0.882	0.275	0.181	0.701	0.172	0.108
ENDSUBG _{InfoNCE}	0.836	0.339	0.192	0.857	0.593	0.219	0.882	0.280	0.196	0.703	0.163	0.105

introduced between nodes in the subset for perturbation from the topological perspective. The detection model is expected to uncover these ASs by considering both attributes and structural properties. Detailed dataset statistics are listed in Table I.

2) *Performance Comparison*: The results with only one AS and three ASs are summarized in Tables IV and V, respectively. We have the following observations.

- 1) ENDSUBG consistently outperforms other baselines in real-world datasets involving different numbers of subgraph anomalies, which strongly confirms the effectiveness of our proposed joint modeling framework for anomaly-aware subgraph partition and anomaly detection. By contrast, AS-GAE relies on a manually set threshold to locate unusual areas, which is difficult to adapt to the unknown data distribution and prone to erroneously narrowing the scope of the abnormal regions. Besides, it treats connected components in the network as extracted candidate subgraphs with the neglect of correlations between graph topology and node attributes. This problematic subgraph division further makes it fail to detect subgraph anomalies accurately (see case study).
- 2) On real-world datasets with no clear subgraph boundaries and more complex types of anomalies, dense block detection models including GAE + DBSCAN and DGI + DBSCAN no longer achieve the same good results as on the synthetic BA graph. From the perspective of AP scores, their performance has declined significantly, and the recognition precision of anomalies is less than 10%. Additionally, the gap between pipeline-based methods and our ENDSUBG once again corroborates the supe-

riority of exploring end-to-end methods to solve the subgraph anomaly detection from a holistic perspective.

3) *Impact of Different Mutual Information Estimators*: We additionally explore the impact of another widely used InfoNCE estimator on detection performance. Since we follow the InfoMax principle [55] to directly maximize the mutual information between input node features and embeddings, which have different dimensions, the InfoNCE formulation in Deep InfoMax [56] is used here to calculate $I(\mathbf{h}_i; \mathbf{x}_j)$

$$I(\mathbf{h}_i; \mathbf{x}_j) = \mathcal{D}_{\theta}(\mathbf{h}_i, \mathbf{x}_j) - \mathbb{E}_{\mathbf{v}_n \sim \mathbb{P}_n} \left[\log \sum e^{\mathcal{D}_{\theta}(\mathbf{h}_i, \mathbf{x}_{v_n})} \right]. \quad (14)$$

We replace only the mutual information estimator and keep other experimental settings unchanged. As can be seen from the last two rows of Tables IV and V, compared with the JSD estimator, InfoNCE achieves both gains and decreases in performance on different datasets. Overall, the performance of these two mutual information estimators is quite comparable, and in practice, they can both be considered promising solutions (with a suitable number of negative samples).

F. Case Studies (Answer to Q4)

We provide intuitive visual analysis to better understand the detected ASs. Since the large number of nodes in real datasets results in poor visualization, we adopt the graph generation strategy mentioned in Section IV-D1 to create a synthetic dataset with the BA model as background graphs. It contains 1050 nodes belonging to ten communities (subgraphs), among which two ASs have denser topological links and deviated node attributes, as shown in Fig. 4. We use different colors

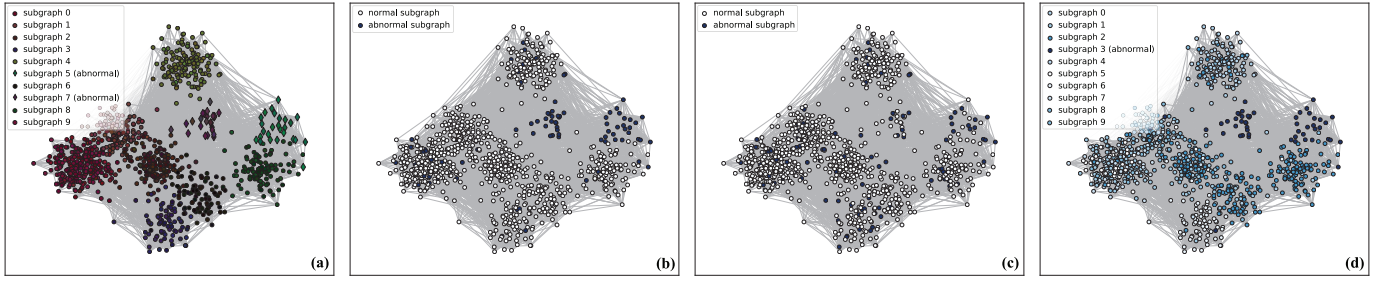


Fig. 4. (a) Synthetic dataset where two ASs are marked by diamond-shaped nodes. Visualization of detection results on the synthetic dataset from (b) DGI + DBSCAN, (c) AS-GAE, and (d) our ENDSUBG.

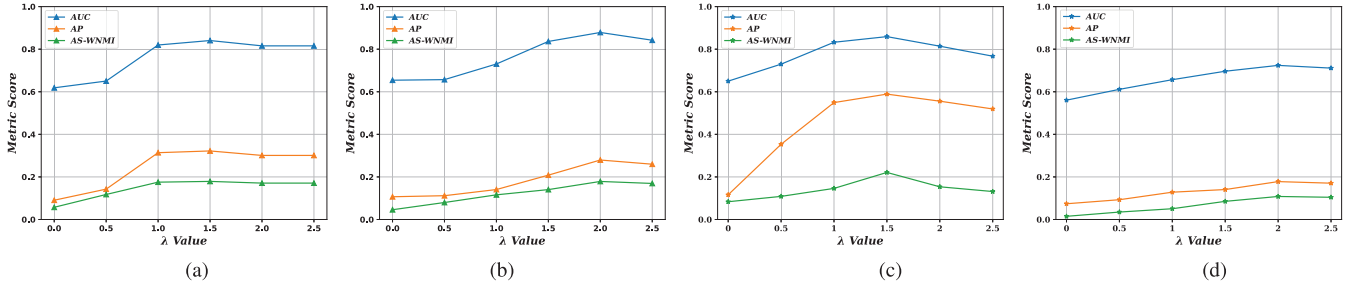


Fig. 5. AUC, AP, and AS-WNMI plots of ENDSUBG with respect to different λ on four real-world datasets. (a) LargeCora 3AS. (b) AmazonComputer 3AS. (c) Reddit 3AS. (d) Questions 3AS.

to display the subgraph division results of each algorithm. For DGI + DBSCAN, although the discovered noisy nodes have a lot of overlap with true ASs, there are still many misidentified anomalies. Besides, DBSCAN only divides the nodes in the input graph into two clusters and does not further distinguish the other eight normal subgraphs in the ground truth, which seems very coarse-grained. As for AS-GAE, the residual graph derived by manual threshold may mistakenly narrow the scope of the anomaly area, and the inappropriately connected subgraph extraction further causes a big gap between the identified AS and the ground truth. By contrast, our ENDSUBG implements explicit anomaly-aware subgraph division, a form of subgraph composition that highlights anomalies in the input graph. The shade of node color represents the anomaly score evaluated by ENDSUBG. The higher the score, the darker the color. It can be seen from Fig. 4(d) that the division of subgraphs is similar to the intrinsic community structure of the input graph. For two ASs, ENDSUBG summarizes them into a unified AS, and almost all the nodes involved are ferreted out. For our assumed normal areas, the model also pre-evaluates the potential risk levels in each subarea. Overall, ENDSUBG shows superior subgraph anomaly detection quality than other baselines.

G. Parameter Analysis

Note that the model ENDSUBG involves several hyperparameters such as the predefined number λ of deviation standards in (1), the number of extracted subgraphs C in (8), an inverse-temperature parameter τ in (7), and the trade-off parameters α and β in (11). In this part, we mainly explore the impact of these important parameters on detection performance.

1) *Extent of Anomaly Perception:* We change λ from 0 to 2.5 in the step of 0.5 and keep the other parameters unchanged. Fig. 5 shows the changes of ENDSUBG regarding the above three metrics on four real-world datasets with three ASs. As can be seen, with the growth of λ , AUC, AP, and AS-WNMI scores all gradually increase and then tend to decline. Through analysis, we attribute it to the fact that when $\lambda = 0$, most edges (e.g., about 59% in LargeCora_3AS and 57% in AmazonComputer_3AS) are treated as potential anomalies and filtered out, including many normal edge structures. This undoubtedly loses a lot of useful information, leading to poor detection performance. As λ grows, the filtering range gradually narrows to real potential abnormal edges, and the performance becomes better accordingly. But when λ is set too large, for example, $\lambda = 2.5$ in AmazonComputer_3AS, all edge links will be preserved and utilized. Then potential anomalies will become noise and negatively impact detection performance, which reflects the importance of anomaly-aware thinking. As for LargeCora_3AS, when λ is set to 2 and 2.5, the model will not filter out any potential anomalies, so the metric values are the same. In practice, we adjust λ to adapt to different data distributions and keep the proportion of filtered edges from being too high as anomalies generally account for a minority. Overall, setting it to 1.5 to 2.0 is more appropriate.

2) *Granularity of Subgraph Partition:* We adjust C from {6, 8, 10, 12, 15, 20, 30} and keep the other parameters unchanged. The AUC, AP, and AS-WNMI of ENDSUBG concerning different C on four real-world datasets are illustrated in Fig. 6. It can be seen that all three metrics improve accordingly as C gradually increases at the beginning. With the continuous rise of C , the performance drops moderately but overall maintains a good and stable trend. Too few subgraph partitions imply oversimplification, which is not conducive to

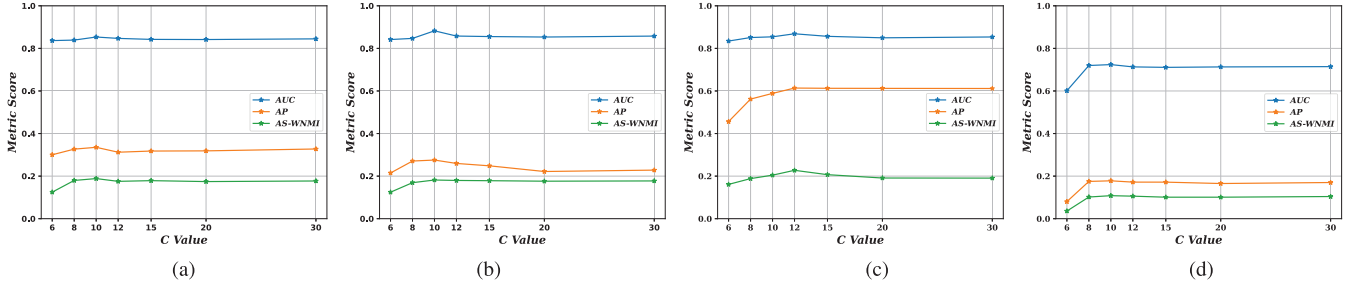


Fig. 6. AUC, AP, and AS-WNMI plots of ENDSUBG with respect to different C on four real-world datasets. (a) LargeCora 3AS. (b) AmazonComputer 3AS. (c) Reddit 3AS. (d) Questions 3AS.

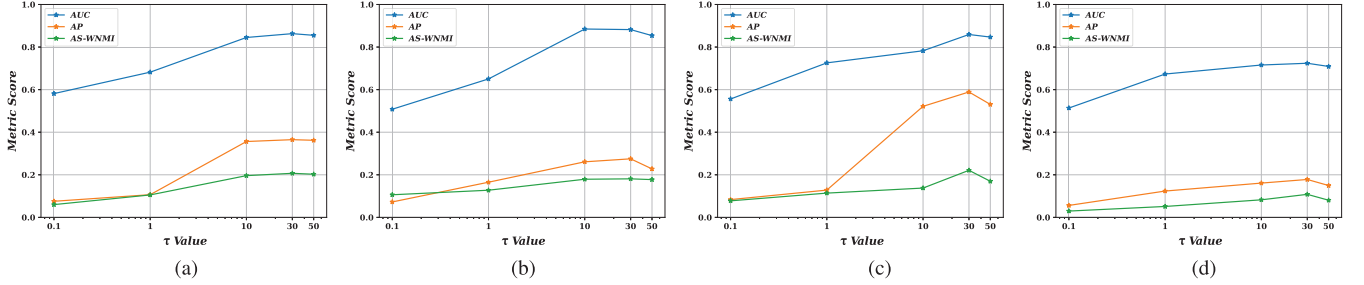


Fig. 7. AUC, AP, and AS-WNMI plots of ENDSUBG with respect to different τ on four real-world datasets. (a) LargeCora 3AS. (b) AmazonComputer 3AS. (c) Reddit 3AS. (d) Questions 3AS.

the discriminator identifying anomalies by learning “subgraph-vicinity” matching patterns. However, if the number of subgraphs is too large, the subgraphs may be too granular and difficult to interpret. The overly subtle distribution differences between subgraphs make it difficult for the discriminator to delineate the boundary between anomalies and benign ones. In practice, we recommend employing unsupervised community detection techniques such as BIGCLAM [57] to automatically estimate the appropriate number of subgraphs in advance.

3) *Impact of Inverse-Temperature Parameter:* We tune τ from $\{0.1, 1, 10, 30, 50\}$ and keep the other parameters unchanged. Fig. 7 shows the changes of ENDSUBG regarding the AUC, AP, and AS-WNMI metrics on four real-world datasets. It can be seen that the detection performance gradually improves as the τ increases. When τ reaches 50, the metrics drop slightly. Recall that in (7), a larger inverse-temperature value will make the probability distribution sharper and tend to give higher membership probabilities to the nearest clusters, which makes the model tend to partition clearer subgraphs. However, an excessive τ value results in overly concentrated output probabilities, weakening the tolerance for overlapping clusters. On the contrary, a smaller temperature value leads to smoother probability distributions, making it more difficult for the model to determine the boundaries of subgraphs and uncover ASs. Overall, the value of τ cannot be set too small in practice, and it seems more appropriate to set it around 10–30.

4) *Effect of Tradeoff Parameters:* We adjust α and β from $\{0.1, 0.3, 0.5, 0.7, 1.0\}$ to explore their effects on detection performance. The performance variance results in terms of AUC, AP, and AS-WNMI on LargeCora_3AS and Amazon-Computer_3AS are shown in Fig. 8. It can be seen that when α and β are both small, all three metrics are at low levels since

poor quality graph embedding and subgraph partition hinder the identification of anomalies. The gradual performance gains come from the increase of α and β , and the promising results fall within the range of α from 0.7 to 1.0 and β from 0.5 to 1.0. Accordingly, we recommend setting both α and β around 0.7–1.0 to ensure a positive foundation for graph embedding and subgraph partition.

H. Computation Load Analysis

In addition to the theoretical complexity analysis, here, we make an intuitive comparison between the computation loads of different methods. Table VI lists the GPU/CPU memory and running time consumed by different methods when detecting datasets containing three ASs. For pipeline methods, we report the highest memory usage among the two stages, as well as the total running time of both stages. Since Louvain + SADE and AMEN are traditional nondeep methods and do not involve any operations on GPU, “/” means not applicable. It can be seen that our method could complete the detection in a relatively short time (although slightly longer on large datasets, but acceptable). Although some methods like AS-GAE and DGI + DBSCAN are somewhat faster, their performance on all three metrics is significantly lower than ours. Since our method involves many complex computational processes, it does consume a lot of GPU/CPU memory, but in exchange for superior subgraph anomaly detection performance. Overall, our ENDSUBG could work normally on commonly used GPUs such as NVIDIA RTX 4090 or A100 and derive promising subgraph anomaly detection results within an acceptable response time. Here, on large dataset Questions_3AS, we employ the sampling strategy to avoid GPU memory overflow. Optimization of the code implementation may further speed up

TABLE VI
COMPUTATION LOADS OF DIFFERENT METHODS WITH RESPECT TO MEMORY ASSUMPTION (MB) AND RUNNING TIME,
WHERE MG DENOTES MAX GPU MEMORY AND MC MEANS MAX CPU MEMORY

Metric	LargeCora_3AS			Reddit_3AS			AmazonComputer_3AS			Questions_3AS		
	MG	MC	Time	MG	MC	Time	MG	MC	Time	MG	MC	Time
ANEMONE	1063.29	1941.91	1918s	456.16	1727.59	1810s	828.82	2802.72	2020s	7969.99	31727.55	3819s
GRADATE	1168.93	18734.70	2423s	892.21	16393.23	2267s	1530.55	26504.72	2611s	15883.04	338624.81	5468s
GAE+DBSCAN	4037.65	1624.26	503s	3570.30	1012.38	572s	5808.41	1759.50	548s	65508.78	18346.02	600s
DGI+DBSCAN	11773.46	660.94	169s	709.38	34.50	188s	3557.97	201.86	222s	905.89	333.06	579s
Louvain+SADE	/	1949.75	175s	/	1737.34	256s	/	2850.13	381s	/	31863.69	3904s
CommDGI+SADE	2043.88	2330.90	2527s	1372.58	2361.84	2889s	2326.19	4373.60	3865s	23969.54	31989.46	24023s
AMEN	/	1637.46	71s	/	1401.90	1429s	/	1462.94	1898s	/	1523.02	784s
AS-GAE	1601.14	1339.16	144s	891.69	433.47	88s	3142.71	696.14	145s	16053.07	7918.99	403s
ENDSUBG	14917.74	3228.79	187s	5730.30	2592.31	213s	9597.61	4222.63	175s	16698.87	47522.53	28539s

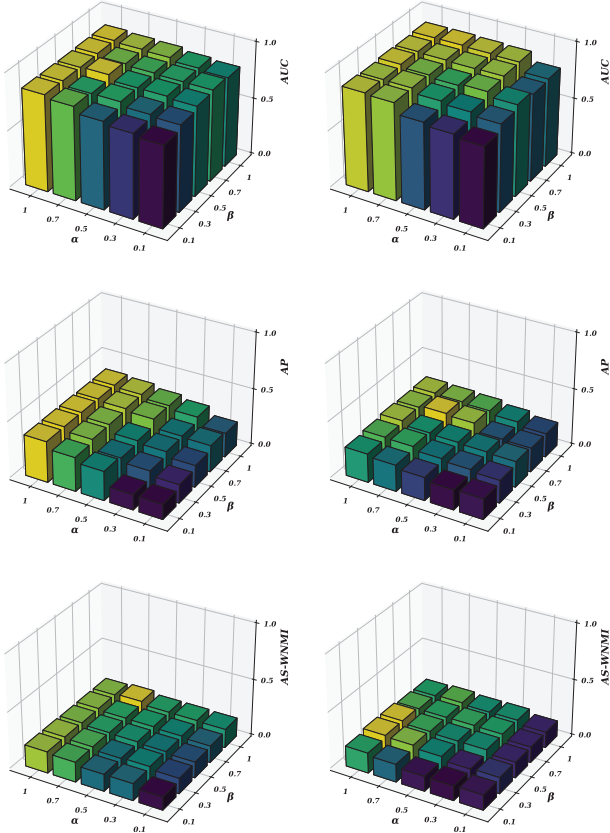


Fig. 8. Effects of tradeoff parameters α and β in terms of AUC, AP, and AS-WNMI on LargeCora_3AS (left) and AmazonComputer_3AS (right).

the operation and reduce memory consumption, which will be left for future research.

I. Further Explorations

Note that real-world scenarios often involve noisy or incomplete data, we further explore the performance of ENDSUBG in the following two situations and illustrate its potential limitations which will be left for future research.

1) *Incomplete Data Scenario*: Here, we simulate different degrees of data missing scenarios by randomly masking 10%, 30%, 50%, and 70% of node attributes. Table VII summarizes the results on two datasets. As can be seen, the model

TABLE VII
PERFORMANCE OF ENDSUBG UNDER DIFFERENT DATA MISSING RATIOS

Mask Ratio	LargeCora_3AS			Reddit_3AS		
	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI
w/o Mask	0.853	0.335	0.188	0.859	0.589	0.221
10%	0.885	0.343	0.189	0.864	0.593	0.228
30%	0.842	0.326	0.154	0.833	0.568	0.182
50%	0.823	0.319	0.152	0.821	0.546	0.173
70%	0.797	0.291	0.146	0.810	0.514	0.170

TABLE VIII
PERFORMANCE OF ENDSUBG UNDER DIFFERENT DATA NOISE

Noise Type	LargeCora_3AS			Reddit_3AS		
	AUC	AP	AS-WNMI	AUC	AP	AS-WNMI
w/o Noise	0.853	0.335	0.188	0.859	0.589	0.221
Attribute Noise	0.825	0.301	0.135	0.819	0.263	0.106
Structural Noise	0.841	0.325	0.201	0.862	0.573	0.219

still works well when a small amount of data is missing (e.g., 10% mask), but as the degree of data incompleteness gradually increases, the performance shows a downward trend. Introducing attribute-missing graph representation learning techniques [58] may be a promising solution to address this limitation.

2) *Noisy Data Scenario*: Here, we consider two types of noise scenarios from the aspects of attributes and topology, respectively. In specific, we add random variables that satisfy the standard normal distribution to the node attributes to generate noisy attributes. In addition, we randomly add 100 edge links in the original topology to simulate structural noise. Table VIII reports the results on two datasets. It can be found that structural noise seems to have less impact on the model, which could benefit from the model's perception and refinement of noise edges that do not conform to the homogeneity assumption. In contrast, the model is not robust enough to attribute noise. Noise-resistant robust graph learning and anomaly detection techniques [59], [60] may be feasible solutions to overcome this limitation.

V. CONCLUSION AND FUTURE WORK

This work concentrates on the subgraph anomaly detection task that has currently received less scrutiny. We present an end-to-end subgraph anomaly detection framework ENDSUBG, which follows the new idea of jointly modeling subgraph partition and anomaly detection rather than treating them as separate steps. ENDSUBG explicitly considers the

potential AS boundaries via anomaly-aware graph embedding and subgraph partition, then spots subgraph anomalies by learning the “subgraph-vicinity” matching patterns on the coarsened subgraph network. To comprehensively evaluate the detection quality from both subgraph partition and anomaly recognition, we propose an AS-WNMI as a new evaluation metric, which fills the gap in performance evaluation for this task. Experiments on both synthetic and real-world datasets validate the superiority of ENDSUBG in discovering suspicious subgraphs with different numbers of subgraph anomalies and diverse background graphs. Future work will concentrate on improving the applicability of the model to practical scenarios with incomplete and noisy data or integrating temporal modules to capture evolutionary information in dynamic applications.

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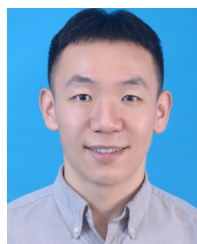
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